

Ziyao (Adonis) Zeng | 曾子尧

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Research Interests

- **Multimodal Learning** inspired by humans for **Agents and Robotics**, especially **VLMs, LLM Agents, Diffusion Models, World Models, 3D Vision**
- My line of work in **"Language for 3D Vision"** explores **how vision-language models can perceive and understand the world like humans do.**

Work Experience

- **Meta Reality Labs** May 2026 - [Expected] Dec. 2026
- Research Scientist Intern | Working with **Dr. Ziyi Kou** and Dr. Li Guan on Dexterous Vision-Language-Tactile Model
- **NVIDIA Research** May 2025 - March 2026
- Research Intern | Worked with **Dr. Hang Su**, Dr. Jindong Jiang, Dr. Abhishek Badki, Dr. Orazio Gallo, and Dr. Sifei Liu
- **Ziyao Zeng et al.** "Boosting **Spatial Understanding in VLMs** with Spatial State Tokens." **U.S. Patent** Application (Pending), NVIDIA Corp., 2026.
- **Shanghai AI Lab** Oct. 2022 - March 2023
- Intern Researcher | Worked on research project of visual grounding and prototype-based segmentation
- **UISEE Technologies** July 2021 - April 2022
- Intern Algorithm Engineer | Developed point cloud recognition & monocular 3D object detection algorithm for self-driving cars

Education

- **Yale University** Aug. 2023 - [Expected] Jan. / May 2027
- **Ph.D. in Computer Science**, advised by **Prof. Yuval Kluger** and **Prof. Daniel Rakita**
- **ShanghaiTech University** Sept. 2019 - June 2023
- **B.Eng. in Computer Science**, Major | Innovation and Entrepreneurship, Minor
- Research Assistant at **PLUS Lab**, ShanghaiTech University; working with **Prof. Xuming He**
- Research Assistant at **GRASP Lab, University of Pennsylvania**; working with **Prof. Jianbo Shi** and **Dr. Renrui Zhang**
- Entrepreneurial Mindshift Program, Babson College, March 2021 | International Summer Program, Osaka University, June 2022

Selected Publications (*order by significance; *: equal contribution*)

- **SciAgentArena: Benchmarking AI Agents for Addressing Scientific Challenges Across Scales** [LLM Agents]
Tianyu Liu*, <6 more co-first authors>, **Ziyao Zeng***, <21 more authors>, Marinka Zitnik, James Zou, Hua Xu, Hongyu Zhao
Under Review of Nature | Link: <https://arxiv.org/abs/2606.12736>
- We introduce **SciAgentArena**, a benchmark with ~200 tasks for evaluating AI agents across diverse **real-world scientific research scenarios**.
- **Iris: Integrating Language into Diffusion-based Monocular Depth Estimation** [VLMs, Diffusion Models, 3D Vision]
Ziyao Zeng*, Jingcheng Ni*, Daniel Wang, Patrick Rim, Younjoon Chung, Fengyu Yang, Byung-Woo Hong, Alex Wong
CVPR 2026 | NECV 2025 Oral Presentation (18.75%) | Link: <https://adonis-galaxy.github.io/Iris-website/>
- We find **language improves diffusion-based depth estimators** through the conditional distributions modeled in text-to-image pre-training.
- **RSA: Resolving Scale Ambiguities in Monocular Depth Estimators through Language Descriptions** [VLMs, 3D Vision]
Ziyao Zeng, Yangchao Wu, Hyoungseob Park, Daniel Wang, Fengyu Yang, Stefano Soatto, Dong Lao, Byung-Woo Hong, Alex Wong
NeurIPS 2024 | Link: <https://arxiv.org/abs/2410.02924>
- The **first** to use language to infer scales of 3D scenes, enabling **zero-shot relative-to-metric depth alignment**.
- **WorDepth: Variational Language Prior for Monocular Depth Estimation** [VLMs, 3D Vision]
Ziyao Zeng, Daniel Wang, Fengyu Yang, Hyoungseob Park, Yangchao Wu, Stefano Soatto, Byung-Woo Hong, Dong Lao, Alex Wong
CVPR 2024 | Link: <https://arxiv.org/abs/2404.03635>
- We formulate **language as a variational prior** by text-VAE, then use **image as condition** to select most likely depth as **posterior**.
- **Can Language Understand Depth?** [VLMs, 3D Vision]
Renrui Zhang*, **Ziyao Zeng***, Ziyu Guo, Yafeng Li
ACM Multimedia 2022 (Brave New Idea, 12.5%) | Link: <https://arxiv.org/abs/2207.01077>
- The **first** to conduct **zero-shot training-free adaptation** from VLMs to monocular depth estimation.
- **UniTac: A Unified Multimodal Model for Cross-Sensor Tactile Understanding and Generation** [VLMs, Multimodal]
Jiahang Tu, Fengyu Yang, Chenyang Ma, Xihang Yu, **Ziyao Zeng**, Shaokai Wu, Hanbin Zhao, Zhi Tao, Chao Zhang, Hui Qian, Alex Wong
ECCV 2026 (Oral) | Link: <https://arxiv.org/abs/2606.31451>
- The **first unified multimodal model** designed for **tactile understanding and generation**.
- **A Physics-Grounded Benchmark for Multi-Agent Dynamics in World Models** [World Models]
Nuo Chen, Lulin Liu, Zihao Li, **Ziyao Zeng**, <6 more authors>, Boris Ivanovic, Marco Pavone, Zhangyang Wang, Yang Zhou, Zhiwen Fan
ECCV 2026 (Oral) | Link: <https://arxiv.org/abs/2606.28757>
- Introduce **CrashTwin**, a **physics-grounded** world model evaluation framework with **25K synthetic** and **12K in-the-wild** collision sequences.
- **RuleSmith: Multi-Agent LLMs for Automated Game Balancing** [LLM Agents]
Ziyao Zeng, Chen Liu, Tianyu Liu, Hao Wang, Xiaotao Sun, Fengyu Yang, Xiaofeng Liu, Zhiwen Fan
Under Review | NE Agents Day 2026 | Link: <https://adonis-galaxy.github.io/RuleSmith-website/>
- Multi-agent LLMs self-play with Bayesian Optimization to automatically balance numerical parameters in asymmetric strategic games.
- **Coffee: Controllable Diffusion Fine-tuning** [Diffusion Models]
Ziyao Zeng, Jingcheng Ni, Ruyi Liu, Alex Wong
Under Review | Link: <https://arxiv.org/abs/2511.14113>
- Coffee controls diffusion fine-tuning to **prevent learning and entangling undesired concepts**.

Other Publications *(order by significance; *: equal contribution)*

- **4DP-QA: Scalable QA for 4D Perception in Vision Language Models** [VLMs, 3D Vision]
Seokju Cho, Abhishek Badki, Hang Su, Jindong Jiang, **Ziyao Zeng**, Seungryong Kim, Sifei Liu, Orazio Gallo
CVPR 2026 | Link: <https://arxiv.org/abs/2606.11568>
- Propose **True-Motion Tracking for VLM** for motion-related scene understanding; construct **400K training data** and a **2.2K-sample benchmark**.
- **Binding Touch to Everything: Learning Unified Multimodal Tactile Representations** [VLMs, Multimodal]
Fengyu Yang*, Chao Feng*, Ziyang Chen*, <3 more authors>, **Ziyao Zeng**, Xien Chen, Rit Gangopadhyay, Andrew Owens, Alex Wong
CVPR 2024 | Link: <https://cfeng16.github.io/UniTouch/>
- Learn a **unified multimodal tactile representation**, enabling touch-LLM, touch-to-image generation, image texture manipulation.
- **PointCLIP V2: Adapting CLIP for Powerful 3D Open-world Learning** [VLMs, 3D Vision]
Xiangyang Zhu*, Renrui Zhang*, Bowei He, **Ziyao Zeng**, Shanghang Zhang, Peng Gao
ICCV 2023 | Link: <https://arxiv.org/abs/2211.11682>
- Adapt CLIP for open-world 3D understanding and exceed PointCLIP by up to **42.90% accuracy**; using **realistic shape projection** and LLMs.
- **OpenLongTail: Generative Scaling of Long-Tail Driving Data** [World Models]
Lulin Liu, <7 more authors>, **Ziyao Zeng**, <4 more authors>, Zhangyang Wang, Weisong Shi, Ruohan Zhang, Marco Pavone, Zhiwen Fan
Under Review | Link: <https://arxiv.org/abs/2607.09655>
- A world model that scales **monocular in-the-wild long-tail driving videos into multi-view assets** for long-tail training.
- **Artificial Foveated Perception for Mitigating Shortcut Learning in Robotic Foundation Models** [VLMs, Robotics]
Xiatao Sun, <6 more authors>, **Ziyao Zeng**, Shawn Li, Qian Wang, Fei Miao, Daniel Rakita
Under Review | Link: <https://arxiv.org/abs/2607.10655>
- Integrates Artificial Foveated Perception into VLA and World Action Model can **suppress spurious scene-level correlations** to improve generalization.
- **ETA: Energy-based Test-time Adaptation for Depth Completion** [3D Vision]
Younjoon Chung, <4 more authors>, **Ziyao Zeng**, Safa Cicek, Byung-Woo Hong, James S. Duncan, Alex Wong
ICCV 2025 | Link: <https://arxiv.org/abs/2508.05989>
- Energy-based test-time adaptation of pretrained depth completion models by quantifying the likelihood.
- **ProtoDepth: Unsupervised Continual Depth Completion with Prototypes** [3D Vision]
Patrick Rim, Hyoungseob Park, S. Gangopadhyay, **Ziyao Zeng**, Younjoon Chung, Alex Wong
CVPR 2025 | Link: <https://arxiv.org/abs/2503.12745>
- A prototype-based approach that learns domain descriptors for continual learning for unsupervised depth completion.
- **iQuery: Instruments as Queries for Audio-Visual Sound Separation** [Multimodal]
Jiabao Chen, Renrui Zhang, Dongze Lian, Jiaqi Yang, **Ziyao Zeng**, Jianbo Shi
CVPR 2023 | Link: <https://arxiv.org/abs/2212.03814>
- Propose Instrument as Query (iQuery) with a flexible query expansion mechanism, achieving up to 44.2 % improvement.
- **HOMER: Homography-Based Efficient Multi-view 3D Object Removal** [3D Vision]
Jingcheng Ni*, Weiguang Zhao*, Daniel Wang, **Ziyao Zeng**, Chenyu You, Alex Wong, Kaizhu Huang
Under review of ACM TOMM | Link: <https://arxiv.org/abs/2501.17636>
- Remove objects across multi-view images with homography-based mapping; propose a new 3D multi-object removal dataset.
- **NeuroBind: Towards Unified Multimodal Representations for Neural Signals** [VLMs, Multimodal]
Fengyu Yang*, Chao Feng*, Daniel Wang*, Tianye Wang, **Ziyao Zeng**, <5 more authors>, Alex Wong
arXiv technical report, 2024 | Link: <https://arxiv.org/abs/2407.14020>
- We propose a **unified neural representation** that aligns multiple neural signal modalities, with pre-trained vision-language embeddings.
- **DSPoint: Dual-scale Point Cloud Recognition with High-frequency Fusion** [3D Vision]
Renrui Zhang*, **Ziyao Zeng***, Ziyu Guo*, Xinben Gao, Kexue Fu, Jianbo Shi
SMC 2023 | Link: <https://arxiv.org/abs/2111.10332>
- Conduct dual-scale processing with local point convolution and global voxel attention with high-frequency coordinate fusion.
- **VT-CLIP: Enhancing Vision-Language Models with Visual-guided Texts** [VLMs]
Longtian Qiu, Renrui Zhang, Ziyu Guo, **Ziyao Zeng**, Yafeng Li, Guangnan Zhang
arXiv technical report, 2021 | Link: <https://arxiv.org/abs/2112.02399>
- Guide text feature to adaptively explore informative regions on the image for CLIP few-shot classification.

Service / Tutoring

Reviewer: CVPR (Outstanding Reviewer 2025), ICCV, ECCV, ICML, ICLR, NeurIPS, ACM MM, AISTATS, ICASSP, BMVC, TCSVT, TMLR;
Volunteer for WWF China & Greenpeace; Yale CS PhD '24 Mentor;
Intern Lecturer at Shanghai Advanced Research Institute 2021
Teaching Assistant for Introduction to Machine Learning, Yale University 2025
Teaching Assistant for Introduction to AI Application, Yale University 2026

Awards / Competitions

Sigma Xi, The Scientific Research Honor Society, Membership Nomination 2025
Merit Student of ShanghaiTech University 2021, 2022
State Grid AI Innovation Competition in Power Dispatch and Control (Top 10%, 24/242) 2021

Skills

VLMs (CLIP, Qwen, InternVL, NVILA), LLM Agents (multi-agent orchestration), Diffusion Models (Stable Diffusion),
World Models (Wan), 3D Vision (monocular depth, point clouds), Robotics (VLA, tactile sensing), Large-scale benchmark & dataset
construction (200-task agent benchmark, 400K VQA); Claude Code, Python/PyTorch, C/C++, Unity/C#